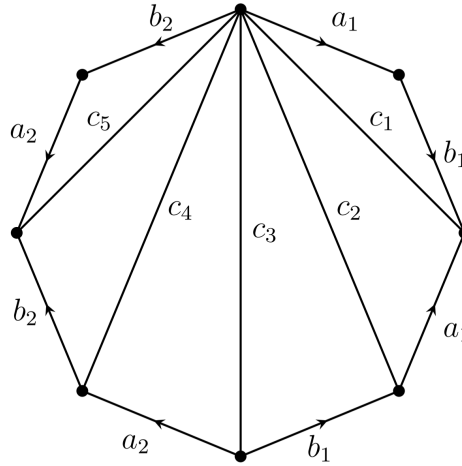


MTH 732 Topology - Homework 6

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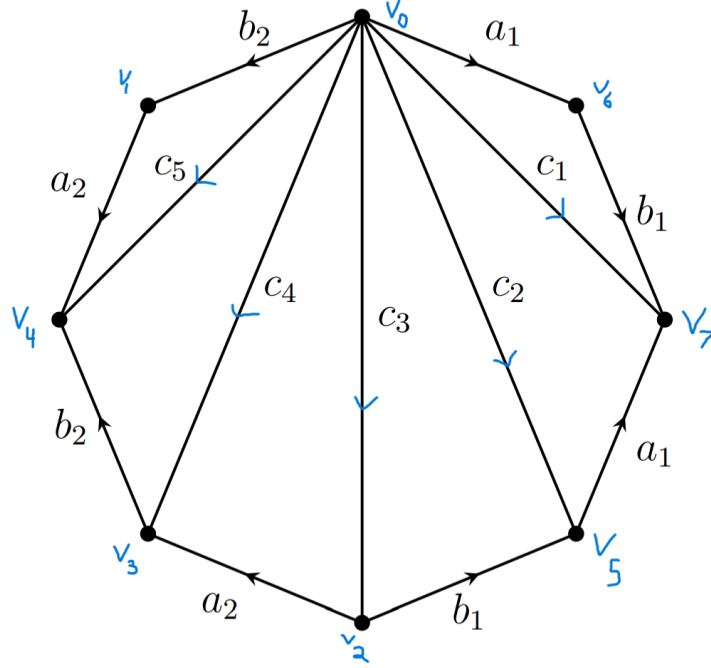
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Assignment 1. The following diagram represents an incomplete Δ -complex structure on the genus 2 surface (2-holed torus). Arrows have been correctly placed on the edges a_i and b_i , but not on the c_i .



- (i) Complete the diagram by orienting the c_i and list all simplices involved (there are 0, 1, and 2-simplices). Keep track of the labels you assign for each simplex.
- (ii) Compute the boundary maps $\partial_2 : \Delta_2 \rightarrow \Delta_1$ and $\partial_1 : \Delta_1 \rightarrow \Delta_0$ explicitly. Use the labels from part (i) to say what the map does to each simplex.
- (iii) Find the homology groups H_0^Δ , H_1^Δ , and H_2^Δ . Check that $H_1^\Delta = \pi_1/[\pi_1, \pi_1]$.

Proof. (i) Our labeling:



Thus the simplices are as follows:

$$0\text{-simplices} = \{[v_0], [v_1], [v_2], [v_3], [v_4], [v_5], [v_6], [v_7]\}$$

$$1\text{-simplices} = \{[v_0, v_1], [v_0, v_4], [v_0, v_3], [v_0, v_2], [v_0, v_5], [v_0, v_7], [v_0, v_6], [v_1, v_4], [v_3, v_4], [v_2, v_3], [v_2, v_5], [v_5, v_7], [v_6, v_7]\}$$

$$2\text{-simplices} = \{[v_0, v_1, v_4], [v_0, v_3, v_4], [v_0, v_2, v_3], [v_0, v_2, v_5], [v_0, v_5, v_7], [v_0, v_6, v_7]\}$$

(ii) First we compute ∂_2 :

$\sigma \in \Delta_2$	$\partial_2(\sigma)$
$[v_0, v_1, v_4]$	$[v_1, v_4] - [v_0, v_4] + [v_0, v_1] = a_2 - c_5 + b_2$
$[v_0, v_3, v_4]$	$[v_3, v_4] - [v_0, v_4] + [v_0, v_3] = b_2 - c_5 + c_4$
$[v_0, v_2, v_3]$	$[v_2, v_3] - [v_0, v_3] + [v_0, v_2] = a_2 - c_4 + c_3$
$[v_0, v_2, v_5]$	$[v_2, v_5] - [v_0, v_5] + [v_0, v_2] = b_1 - c_2 + c_3$
$[v_0, v_5, v_7]$	$[v_5, v_7] - [v_0, v_7] + [v_0, v_5] = a_1 - c_1 + c_2$
$[v_0, v_6, v_7]$	$[v_6, v_7] - [v_0, v_7] + [v_0, v_6] = b_1 - c_7 + a_1$

and then ∂_1 :

$\sigma \in \Delta_1$	$\partial_1(\sigma)$
$[v_0, v_1]$	$[v_1] - [v_0]$
$[v_0, v_4]$	$[v_4] - [v_0]$
$[v_0, v_3]$	$[v_3] - [v_0]$
$[v_0, v_2]$	$[v_2] - [v_0]$
$[v_0, v_5]$	$[v_5] - [v_0]$
$[v_0, v_7]$	$[v_7] - [v_0]$
$[v_0, v_6]$	$[v_6] - [v_0]$
$[v_1, v_4]$	$[v_4] - [v_1]$
$[v_3, v_4]$	$[v_4] - [v_3]$
$[v_2, v_3]$	$[v_3] - [v_2]$
$[v_2, v_5]$	$[v_5] - [v_2]$
$[v_5, v_7]$	$[v_7] - [v_5]$
$[v_6, v_7]$	$[v_7] - [v_6]$

- (iii) First, since $\partial_0 = \mathbf{0}$, $\ker \partial_0 = \Delta_0$, and since every 1-simplex $[v_i]$ where $i \neq 0$ satisfies $[v_i] - [v_0] \in \text{im } \partial_1$, they all get identified in H_0^Δ , so $H_0^\Delta = \mathbb{Z}$.

In H_1^Δ , we have the c_i edges identified as follows:

$$\begin{aligned}
c_5 &= a_2 + b_2 \\
c_4 &= c_5 - b_2 = a_2 \\
c_3 &= c_4 - a_2 = 0 \\
c_2 &= b_1 + c_3 = b_1 \\
c_1 &= a_1 + c_2 = a_1 + b_1,
\end{aligned}$$

so we can in fact express each c_i in terms of a_1 , a_2 , b_1 , and b_2 , leaving us with four generators for H_1^Δ . Thus $H_1^\Delta \cong \mathbb{Z}^4$.

Finally, consider that Δ_3 is trivial, so $\text{im } \partial_3$ is trivial, and thus $H_2^\Delta = \ker \partial_2$. Since there are 6 generators in Δ_2 and the rank of $\text{im } \partial_2$ is 5 as seen above, $\ker \partial_2$ must have rank 1, and thus $H_2^\Delta \cong \mathbb{Z}$.

□

Assignment 2. Show that if $X = \sqcup_{i \in I} X_i$ where the X_i are the path components of X , then for every n , the Abelian groups $C_n(X)$ and $H_n(X)$ split as direct sums:

$$C_n(X) = \bigoplus_{i \in I} C_n(X_i) \quad \text{and} \quad H_n(X) = \bigoplus_{i \in I} H_n(X_i).$$

Proof. Since each singular n -simplex is a continuous map, it must map into a single path component X_i . Thus the free abelian group on all singular n -simplices, namely $C_n(X)$, is the direct sum of the free abelian groups on all singular n -simplices living in each X_i separately, namely $\bigoplus_{i \in I} C_n(X_i)$.

Given the above, the kernels of the boundary maps will also split since they are subgroups of the $C_n(X_i)$, and thus we can focus on the images of boundary maps. Given a singular

n -simplex σ , the boundary $\partial\sigma$ is a $\mathbb{Z}$ -linear combination of singular $(n-1)$ -simplices, the images of which are subsets of the image of σ , which we have already noted is a subset of X_i . A general element of $H_n(X)$ is a coset $A(\text{im } \partial_{n+1})$ where $A \in C_n(X)$ and thus $A = \sum_{i=1}^m A_i$ where each $A_i \in C_n(X_i)$, and thus $A(\text{im } \partial_{n+1}) = \sum_{i=1}^m A_i(\text{im } \partial_{n+1})$, so $H_n(X) = \bigoplus_{i \in I} H_n(X_i)$. $\square$

Assignment 3. Show that if X is path-connected, $H_0(X) = \mathbb{Z}$.

Proof. For any space X , $\ker \partial_0 = C_0(X)$, so $H_0(X) = C_0(X)/\text{im } \partial_1$. If X is path-connected, then given any two points $x, y \in X$, $x - y \in \text{im } \partial_1$ since there exists a map from a closed interval (which is itself a singular 1-simplex) to X which starts at y and ends at x , and the boundary of that singular 1-simplex (after precomposition with the canonical map from the singular 1-simplex so that it is truly a singular 1-simplex) is $x - y$. Thus by definition of the quotient, since $x - y \in \text{im } \partial_1$, $x \sim y$, and thus every point is related to every other point. Since $C_0(X)$ is generated by the collection of points in X , this collapses the group to one generator. We exhibit the map $\varepsilon : C_0(X) \rightarrow \mathbb{Z}$ defined by

$$\varepsilon \left(\sum_{i=1}^m n_i \sigma_i \right) = \sum_{i=1}^m n_i,$$

which surjects onto $\mathbb{Z}$, and thus $H_0(X) \cong \mathbb{Z}$. $\square$